# Improving Cantonese translation in existing multilingual translation models

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## Abstract

Cantonese is a low-resource language, i.e. it has received a very low amount of attention in NLP research compared to the number of speakers. Recently, high-quality datasets such as SMOL and GATITOS have been released, providing professionally translated parallel data on multiple levels (token-level, sentence-level, and document-level) for low-resource languages like Cantonese. This project explores how to utilize these datasets to improve the performance of a pre-trained multilingual translation model, MADLAD, on Cantonese to English translation tasks. For the token-level data we try to exploit the similarity to written Mandarin, on which models perform much better, by linearly transforming Cantonese embeddings to MADLAD's Mandarin embedding space, but this made no improvement in practice. For the sentence- and document-level data we show that using it to finetune MADLAD through backtranslation significantly improves model performance (6.46% improvement in BLEURT score).

## 1 Introduction

This project explores improving machine translation tasks working with Cantonese text. Although Cantonese has more than 85 million speakers, it is very underrepresented in machine learning and translation, and large training corpusses are rare, as the language is most used in its spoken form. This results in many Cantonese NLP tasks simply being handled by models trained mostly or exclusively on Mandarin language data.

In this project, we explore how both unsupervised and supervised learning can be applied to improve the quality of translation tasks. For this, we use the professionally translated data in the SMOL[1] and GATITOS[3] datasets and the unlabeled data in the dataset collected by Dare et al.[2].

The first experiment performed was a simple token replacement. As there is overlap between Cantonese and Mandarin characters, we replace the tokens that are different in the Cantonese input with their Mandarin equivalents before feeding the text to a pre-trained MADLAD[5]. This simple approach showed some promise, but was limited by not being able to handle differences in grammar and differences in meaning of characters that are the same in both languages.

We use the unlabeled data to train Cantonese token embeddings and then transform these embeddings into the embedding space of a pre-trained English-Mandarin model, MADLAD, using the token-level translations in GATITOS. This did not work in practice, probably because our approach to learning the Cantonese embeddings did not learn the same linear relationships between tokens as the embeddings that MADLAD learnt with masked language modelling.

We then finetune MADLAD on a combination of SMOLSENT and SMOLDOC and evaluate the performance of the finetuned model on a held-out test set from SMOLSENT. This has very good results and improves the test set BLEURT mean by $0.044$ (improvement of $6.46\%$) in our case.

# 2 Related works

## 2.1 MADLAD

MADLAD As a base for our experiments, and later as a model to finetune, we use the MADLAD model. This is a general translation model, with 10.7B parameters, trained on web-scraped unlabeled data in 419 languages. The dataset contains both Mandarin and Cantonese data, but the amount of Cantonese data is very limited compared to the amount of Mandarin data (approximately 61 times less). The model architecture is a standard transformer-based sequence-to-sequence model, similar to the architecture used in models such as T5[9].

# 3 Datasets

## 3.1 SMOL

SMOL[1] (Set of Maximal Leverage) is a high-quality translation corpus from English to more than a hundred low-resource languages. SMOL includes SMOLSENT and SMOLDOC, which represent sentence-level and document-level translations, and they are accompanied by Gatitos[3] which provides token-level translations across many of the same languages. In this project specifically, we use the English-Cantonese portion of SMOL and GATITOS, where there are translations on all three levels of data. For use in our experiments, we split the SMOLSENT dataset such that there is a 80/10/10 split for training/validation/test. We make sure that all splits are completely distinct and are consistent across experiments. We only use training and validation during our experiments, saving the test set for a final evaluation of our models.

## 3.2 Monolingual Cantonese corpus by M. Dare, et. al

In their work on unsupervised Mandarin-Cantonese machine translation, Dare et. al.[2] collected a large corpus of monolingual Cantonese sentences from various online sources. In total 912258 lines were collected by scraping and cleaning data from Wikipedia, Instagram, and a few other sites including a few corpusses compiled for previous academic work. Common for most sources that form the corpus is that the content is mostly user-generated text, which often includes non-standard spellings and colloquial language, which is typical for Cantonese as a language. We use this corpus to train Cantonese token embeddings for the embedding transform experiment described in Section 3.

# 4 Baseline

We evaluate our model on the SMOL validation set using BLEURT. This gives us a BLEURT score of 0.689, by plotting the BLEURT scores on a hexbin plot using bins over the distribution of Cantonese-only characters (see figure 1), we can see that the model seems to sometimes perform significantly worse when there are Cantonese-only characters in the sentence. This can be seen more clearly in a violin plot, see figure 2. TODO: validation plot doesnt really show this :-(

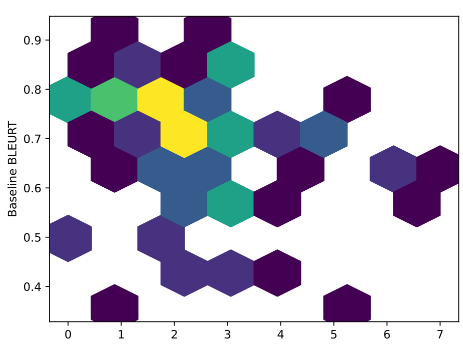


Figure 1: Hexbin plot of the BLEURT scores of the baseline model on the validation set.

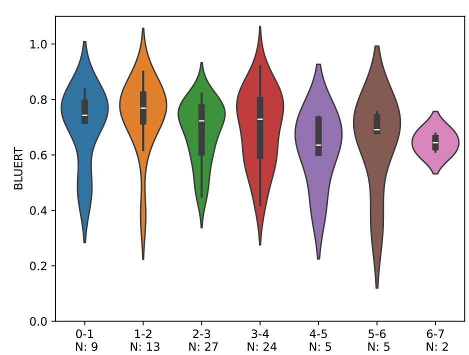


Figure 2: Violin plot of the BLEURT scores of the baseline model on the validation set.

# 5 Experiments

## 5.1 Token replacement

Since the MADLAD model appears to have difficulty with sentences that contain words that are native to written Cantonese, we do an experiment where we replace words that are native to written Cantonese with their Mandarin Chinese translations. To find words that are native to written Cantonese we use CantoDict[10], a collaborative project where Cantonese speakers annotate Cantonese characters with information about their meaning, pronounciation and usage in written Cantonese and Mandarin. If a character in a sentence is used in Cantonese but not used in Mandarin, we try to find its translation in the Gatitos dataset and do a token-level replacement as preprocessing to MADLAD inference. This is a very naive approach, as we do not take context into account when replacing tokens, and when a character has multiple possible translations we simply choose the first one listed in Gatitos. This can also be seen in the results, listed in section 6.1.

## 5.2 Embedding transform

From the promising results of the token replacement experiment 6.1, we believe that giving the model more knowledge of Cantonese characters could significantly improve its Cantonese translation ability.

If we were to simply train a model on GATITOS token-level data it might get slightly better at some characters in the small GATITOS dataset, but we would not be able to generalize the model to become better at a significant amount of Cantonese vocabulary.

Instead we propose the following experiment: We train embeddings on a large corpus of unlabeled Cantonese data that is higher quality than the data MADLAD was trained on, hoping to get embeddings that better represent semantic similarity between Cantonese tokens than the embeddings of MADLAD do. We then perform an embedding transform from our Cantonese embeddings to MADLAD's Chinese embedding space using token-level translations from GATITOS, copying the approach by Mikolov, et. al[7]. Specifically, we train a transformation matrix $W$ such that for a specific Cantonese-Chinese token embedding pair $E_{canto_i}, E_{mandarin_i}$ the embedding transformation $E_{canto_i}W$ approximates $E_{mandarin_i}$, by minimizing the following loss over $W$ with the ADAM-algorithm[4].

$$\sum_{i=1}^{n} ||E_{canto_i}W - E_{mandarin_i}||^2$$

GATITOS is a many-to-many dataset of token-level translations, so for each one-to-many relation we put the target Mandarin embedding to be the average of the target token's embedding.

For training the Cantonese embeddings we use the Monolingual Cantonese corpus by M. Dare, et. al. to train both a Word2Vec[6] model and a GloVe[8] model, where we separate words using the MADLAD tokenizer.

## 5.3 SMOL finetuning

With the disappointing embedding transform results (6.2), we hope to instead improve the models understanding of Cantonese by finetuning it on SMOL Cantonese->English backtranslation. With only the SMOLSENT training set we do not have that much data, so we combine it with the SMOLDOC dataset by flattening the SMOLDOC sentence translation pairs. This approach gives us about 10x more data, while lowering the quality of the data slightly.

# 6 Results

## 6.1 Token replacement

After performing token replacement and translating the resulting sentences with MADLAD, we calculate BLEURT scores. On average, the token replacement approach performs slightly worse than the baseline, with an average BLEURT score of 0.6704 compared to the baseline's 0.6807. We do however observe that the translation of many sentences improve significantly, both by BLEURT score and by our personally determined translation quality. At the same time, many sentences also see a significant drop in BLEURT score and translation quality. A few examples of this can be seen in the table **??**. To check whether the ratio of Cantonese characters in the source sentence has an effect on the performance of the token replacement approach, we bin the sentences by their ratio of replaced characters and plot violin plots of the BLEURT scores for each bin in figure 3. We see no real trend in the results, with both the baseline and token replacement approach performing similarly across all bins, although there is a slight tendency for the token replacement approach to perform better on sentences with a higher ratio of Cantonese characters.

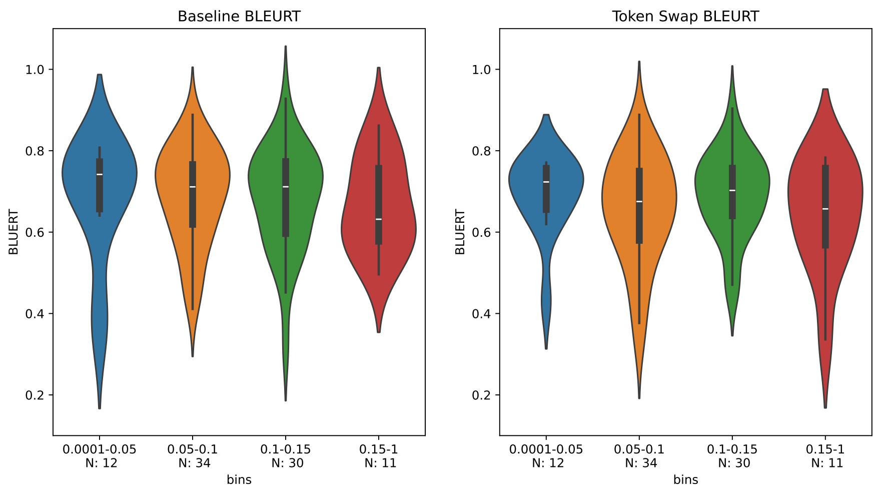


Figure 3: BLEURT scores for token replacement experiment, binned by ratio of Cantonese-only characters in the source sentence.

| Source | Baseline | Token Replacement | Δ BLEURT |
|---|---|---|---|
| Rohan was my sibling who always flew a kite at noon, even if it was cloudy. | The bridge is opened to the public at noon, and closed at midnight. | Rohan is my brother, and he flies kites at noon every day, even when the sky is dark. | $0.449$ |
| From 10 feet away you could tell the bottom of the boat was in rough shape. | It was very bad from 10 feet away. | The bottom was in poor condition and visible from a distance of 10 feet; | $0.246$ |
| It includes a legend of a devil beast, a century-old family curse. | It includes a legendary demon beast, a family curse that has lasted for hundreds of years. | He is the one who has been the master of the witchcraft for centuries. | $-0.380$ |
| Bear in mind, constantly try to purchase silk plants, as they tend to appear like the genuine thing. | Remember, keep trying to buy silk plants because they look like real ones. | They are often used to make the plant's leaves and flowers look like they are actually flowers. | $-0.291$ |

Figure 4

### 6.2 Embedding transform

We train the GloVe embeddings using Stanford's GloVe implementation[1] for 15 epochs. We train the Word2Vec embeddings using Gensim's Word2Vec implementation[2] for 6 epochs.

We train the embedding transformation matrix using stochastic gradient descent with a learning rate of $10^{-2}$ and a batch size of 64 for 175 epochs until it stops improving. A loss plot for training the transformation matrix for our GloVe embeddings can be seen on figure 5.

[1] https://nlp.stanford.edu/projects/glove/

[2] https://radimrehurek.com/gensim/models/word2vec.html

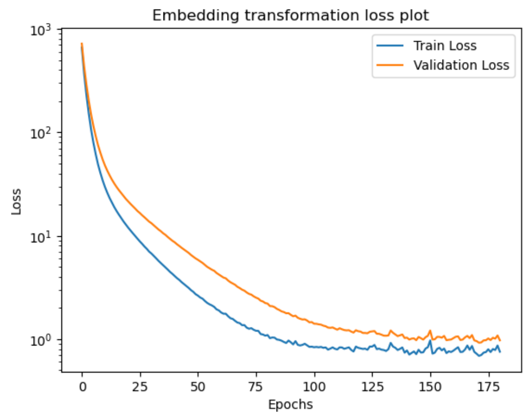


Figure 5: Plot of GloVe transformation matrix loss during training

The transformed embeddings for both Word2Vec and GloVe are however quite bad and generate seemingly random output, although it is still mostly coherent speech. This leads us to believe that there is no approximate linear transformation between the Word2Vec / GloVe embeddings and the MADLAD embeddings which have been trained through masked language modeling. Here are two example outputs:

Original:

As a reminder, this company launched a 34-inch curved gaming monitor last October.

Transformed:

The company is now in the process of launching a new product, the 10-month-old baby-size baby-sized baby-sized baby-sized baby-sized baby-sized baby-sized baby-sized baby-sized baby-sized baby-sized baby-sized baby-sized

Original:

Rohan was my sibling who always flew a kite at noon, even if it was cloudy.

Transformed:

I'm a little bit shy, but I'm not alone.

### 6.3 SMOL finetuning

We finetune with a learning rate of $5 \cdot 10^{-5}$ and a batch size of $8$ with a weight decay of $0.01$ using a A100 GPU on Google Colab. After $1$ epoch ($784$ steps) the model starts to overfit, and no longer improves on the test set, as can be seen on figure **??**, so we use a checkpoint of the model saved after the first epoch.

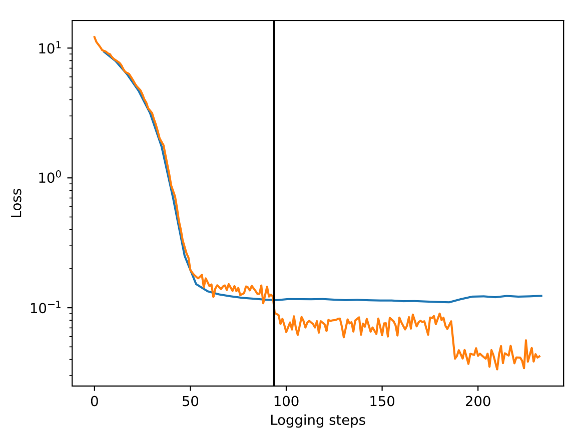


Figure 6: Training loss during finetuning on SMOLSENT + SMOLDOC. The vertical line indicates the end of the first epoch where training was stopped.

| 95% CI | Baseline | Token swap | **Finetuned** | Finetuned TS |
|---|---|---|---|---|
| Test set BLEURT | $[0.6544, 0.7069]$ | $[0.6451, 0.6958]$ | $[0.7059, 0.7449]$ | $[0.6948, 0.7340]$ |

Figure 7: Confidence interval for BLEURT Scores on the test set. We calculate the confidence intervals with the $t$ distribution with degrees of freedom equal to the length of our test set minus one. We use Bessel's correction for the sample standard deviation.

# 7 Discussion

## 7.1 Limitations

Since the GATITOS dataset consists of many-to-many translations, and in some cases there is loss of information (for example translation from a gendered pronoun to non-gendered pronoun). Thus our naive approach of taking all translations from GATITOS may have an adverse effect on performance.

The embedding transform experiment had bad results, we believe that in addition to the above point this was because the linear relations between vocabulary in embedding space learnt by our models (Word2Vec & GloVe) differ from the linear relations between vocabulary in MADLAD's embeddings, due to MADLAD learning its embeddings through a significantly different method (masked language modelling). We therefore also believe that this approach could have had better results if we had pretrained our embeddings in the same way as MADLAD did, using masked language modelling on a T5 model. Unfortunately in this project we do not believe we have the required resources to perform such an requirement.

For multiple of our experiments, we believe that the limited size of the GATITOS and SMOL datasets have been the main bottleneck for achieving better performance. For example, in the token swap experiment, see figure 3, where the sizes of the bins have been listed. We see that in total our test set only contains 87 sentences. For a NLP task, where the input space is very large, this is a very small number of samples to draw conclusions from.

Both our training dataset and our test datasets are backtranslation datasets, we therefore have not measured the proficiency of our model to do translation in the original direction.

## 7.2 Other methods

We could also have performed non-linear transformations of the embeddings, for example by using ReLU or sigmoid activations in a mutlilayer neural network. We did not do this because the point of the embedding transformation is to keep all the linear relations between our vocabulary in embedding space. Furthermore, by increasing model complexity we will very likely to overfit as there will be significantly more parameters in our model than datapoints in the training set.

## 7.3 Conclusion

We have shown a way of utilizing specific aspects of the SMOL dataset (sentence-level and document-level) to improve a trained models translation ability, resulting in an improvement in mean BLEURT score of 0.044 (6.46%). We have also proposed a way of improving the embedding of a trained models using token-level data, and although we were not able to achieve an improvement with this approach in this project, we have proposed a higher-resource method that could work.

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